Fiscal Year:	FY 2009	**Task Last Updated:**	FY 03/12/2009
PI Name:	Bloomberg, Jacob J. Ph.D.		
Project Title:	Development of a Gait Adaptability Training Program as a Countermeasure for Postflight Locomotor Dysfunction		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	10/01/2004	**End Date:**	10/01/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/1/2008 because of reporting requirements (3/09 jvp)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cohen, Helen (Baylor College of Medicine) Mulavara, Ajitkumar (USRA)		
Grant/Contract No.:	NCC 9-58-NA00403		
Performance Goal No.:			
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Task Description:	During the acute phase of adaptation to novel gravitational environments on exploration class missions sensorimotor disturbances have the potential to disrupt the ability of astronauts to ambulate and perform functional tasks. The goal of this project was to develop a sensorimotor adaptability (SA) training program designed to facilitate recovery of locomotor and functional capabilities when astronauts transition to different gravitational environments and during the postlanding recovery period following return to Earth. We have previously shown that training with task variability along with exposure to a variety of surrogate sensorimotor challenges leads to faster adaptation to new environments that have not been previously explored. By applying these concepts for training astronauts we can enhance their ability to "learn how to learn" to adapt to new gravitational environments. To achieve the project's goal we applied these adaptability motor-learning concepts to develop an astronaut SA training program. The project conducted a series of studies that investigated the efficacy of treadmill training combined with a variety of sensory challenges designed to increase adaptability including alterations in visual flow, body loading, and support surface stability. The main findings were: SA training using a treadmill (an analogue to overground locomotion) was effective in producing increased adaptability in a more complex over-ground ambulatory task on an obstacle course. This confirms that for a complex task like walking, treadmill training contains enough of the critical features of overground walking to be an effective training modality. Balance board training alone did not increase training efficacy demonstrating the limitations of training by showing that the training tasks must have certain critical features to allow full generalization. Training that used constantly varying and challenging sensory input during each training session gave subjects a greater ability to rapidly reorganize appropriate response strategies when encountering a novel sensory environment and did not produce interference to motor learning. This indicates that training modes can be designed to maximally promote the rapid adaptability that would be required to deal with an emergency scenario requiring immediate responses soon after landing on a planetary surface. To investigate the utility of body unloading as a training modality we demonstrated that walking on a treadmill for 30 minutes while supported at 40% of body weight via a pneumatic harness produced adaptive alteration in head movement control, lower limb kinematics, and gait cycle timing. These results indicate that alteration in body load can serve as an effective adaptability training modality. Using a treadmill mounted on top of a six degree-of-freedom motion base platform we investigated locomotor training responses produced by subjects introduced to a dynamic walking surface. This study examined the strategies employed by subjects exposed to continuous, sinusoidal lateral motion of the support surface while walking on a treadmill. The results indicate that in order to optimize gait stability, some subjects depending more heavily on vision and others on proprioception. These results allow us to better design training modalities specifically targeted to an individual crewmember's sensorimotor biases. In an associated study we confirmed that SA training could be used to improve balance and locomotor performance in older adults and points to the general applicability of this type of training in different populations. The data obtained in this project will aid in the design of a countermeasure system used to support exploration-class space missions. The operational version of this countermeasure will provide sensory challenges during regular exercise in both pre-and inflight training modes to allow crewmembers to maximize their motor response adaptability facilitating the adaptive transition to partial or unit gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	From a clinical perspective exposure to varied sensorimotor stimuli during treadmill walking provides environmental enrichment to enhance gait and postural adaptability. A vast literature demonstrates that the central nervous system requires a rich and varied sensorimotor environment to maintain normal structure and function. Also, a converging body of evidence supports the use of environmental enrichment both before and after brain trauma, as an intervention to help restore behavioral function These studies indicate that exposure to environmental enrichment can play a central role in the restoration of functional mobility. Importantly, studies have shown that training comprised of challenging balance tasks compared to simple repetitive or treadmill exercises alone increase structural plasticity in the motor cortex and thalamus. Within the context of locomotor functional training and rehabilitation, it is clear that an enriched and challenging environment provided by virtual reality devices provides the optimal conditions to explore variability and in that exploration increase adaptability of motor function. Indeed, the concept of employing virtual reality technology as a tool for rehabilitation has become a topic of great interest among a number of researchers. A collateral benefit of the application virtual reality technology, in this context, will be to make gait training programs for both space flight and rehabilitation more interesting, ultimately leading to increased adherence to prescribed training regimens. In addition, the gait adaptability training program being developed by this project can be used to reduce the propensity of falling in the elderly. We have pursued this line of research in collaboration with Drs. Ronita Cromwell and Regina Buccello-Stout at the University of Texas Medical Branch in Galveston. This associated study compared a group of elders receiving the specialized treadmill adaptability training developed under NSBRI funding with a group that received the treadmill training only. Results from this study showed significantly improved gait function in elders that received the sensorimotor adaptability training (Buccello-Stout et al., 2008) . This study clearly demonstrated the general utility of the training method beyond the NASA applications. Buccello-Stout, RR, Bloomberg, JJ, Cohen, HS, Whorton, EB, Weaver, GD, & Cromwell, RL. Effects of sensorimotor adaptation training on functional mobility in older adults. J Gerontol B Psychol Sci Soc Sci. 63(5): 295-300. 2008.

Task Progress:	In the final year of this study we examined the role of support surface instability as a training modality. Using a treadmill mounted on top of a six degree-of-freedom motion base platform we investigated locomotor training responses produced by subjects introduced to a dynamic walking surface. This study examined the strategies employed by subjects exposed to continuous, sinusoidal lateral motion of the support surface while walking on a treadmill. Subjects tended to either fix themselves in space (FIS), allowing the treadbelt to move laterally beneath them, or fix themselves to the base (FTB), moving laterally as the motion base oscillated (Brady et al., Gait & Posture, In press, 2009). This finding suggests that in order to optimize gait stability, some subjects depending more heavily on vision (FIS group) and others on proprioception (FTB group). These results allow us to better design training modalities targeted to individual crewmember's sensorimotor biases. Recent evidence suggests that sensorimotor adaptability training effects can be retained for months. Given the in-flight logistic constraints on hardware volume and power as well as crew time, preflight training without or with a limited in-flight component would be a preferable training strategy. To minimize cost and demands on crew time integrating sensorimotor adaptability training with existing exercise activities would also be preferable. To address these requirements we have pursued the following research paths: In this final year of the project, a study has been underway to investigate the long-term retention characteristics (months) of sensorimotor adaptability training, using combined visual flow/unstable support surface training. Results from this study will allow us to design preflight sensorimotor adaptability training programs that can be implemented preflight and then retained for many months during expedition class mission. We have also been working with the HRP Musculoskeletal Alterations team, led by Dr. T. Lang. The balance training methods developed in the current project are being integrated into a single countermeasure system that will combine resistance, aerobic, and balance training modalities. During this year the project produced 5 presentations and 3 publications in peer reviewed journals.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Articles in Peer-reviewed Journals	Brady RA, Peters BT, Bloomberg JJ. "Strategies of healthy adults walking on a laterally oscillating treadmill." Gait Posture. 2009 Feb 20. [Epub ahead of print] PMID: 19233653 , Feb-2009
Articles in Peer-reviewed Journals	Buccello-Stout RR, Bloomberg JJ, Cohen HS, Whorton EB, Weaver GD, Cromwell RL. "Effects of sensorimotor adaptation training on functional mobility in older adults." J Gerontol B Psychol Sci Soc Sci. 2008 Sep;63(5):P295-300. PMID: 18818444 , Sep-2008
Articles in Peer-reviewed Journals	Cohen HS, Bloomberg JJ. "Modified dynamic visual acuity tests after acoustic neuroma resection." Acta Otolaryngol. 2007 Aug;127(8):825-8. PMID: 17762992 , Aug-2007
Articles in Peer-reviewed Journals	Moore ST, MacDougall HG, Peters BT, Bloomberg JJ, Curthoys IS, Cohen HS. "Modeling locomotor dysfunction following spaceflight with Galvanic vestibular stimulation." Exp Brain Res. 2006 Oct;174(4):647-59. PMID: 16763834 , Oct-2006
Articles in Peer-reviewed Journals	Mulavara AP, Cohen HS, Bloomberg JJ. "Critical features of training that facilitate adaptive generalization of over ground locomotion." Gait Posture. 2009 Feb;29(2):242-8. Epub 2008 Oct 5. PMID: 18838271 , Feb-2009
Articles in Peer-reviewed Journals	Mulavara AP, Houser J, Miller C, Bloomberg JJ. "Full-body gaze control mechanisms elicited during locomotion: effects of VOR adaptation." J Vestib Res. 2005;15(5-6):279-89. PMID: 16614474 , Oct-2005
Articles in Peer-reviewed Journals	Peters BT, Bloomberg JJ. "Dynamic visual acuity using “far“ and “near“ targets." Acta Otolaryngol. 2005 Apr;125(4):353-7. PMID: 15823804 , Apr-2005
Articles in Peer-reviewed Journals	Mulavara AP, Richards JT, Ruttley T, Marshburn A, Nomura Y, Bloomberg JJ. "Exposure to a rotating virtual environment during treadmill locomotion causes adaptation in heading direction." Exp Brain Res. 2005 Oct;166(2):210-9. PMID: 16034569 , Oct-2005
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Articles in Peer-reviewed Journals	Richards JT, Mulavara AP, Bloomberg JJ. "The interplay between strategic and adaptive control mechanisms in plastic recalibration of locomotor function." Exp Brain Res. 2007 Apr;178(3):326-38. PMID: 17061092 , Apr-2007